8 **(a)** A cell of e.m.f. E and internal resistance r is connected in series to a resistor of resistance R.

(i) Define *electromotive force*.

..... [1]

(ii) Show, by considering energy conversion, that V the terminal potential difference of the cell is

$$V = E - Ir$$

where I is the current flowing in the cell.

[2]

(b) Fig. 8.1 shows a battery with e.m.f. E and an internal resistance r connected to a uniform nichrome resistance wire MN. J is a movable jockey which can slide along wire MN. The voltmeter and the ammeter are taken to be ideal.

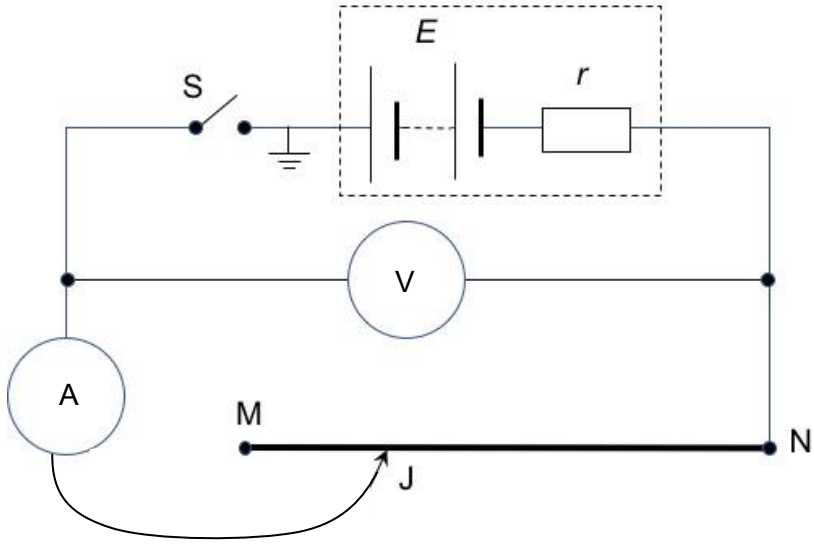

Fig. 8.1

The voltmeter readings V and ammeter readings I obtained for different lengths of JN are used to plot the graph in Fig. 8.2.

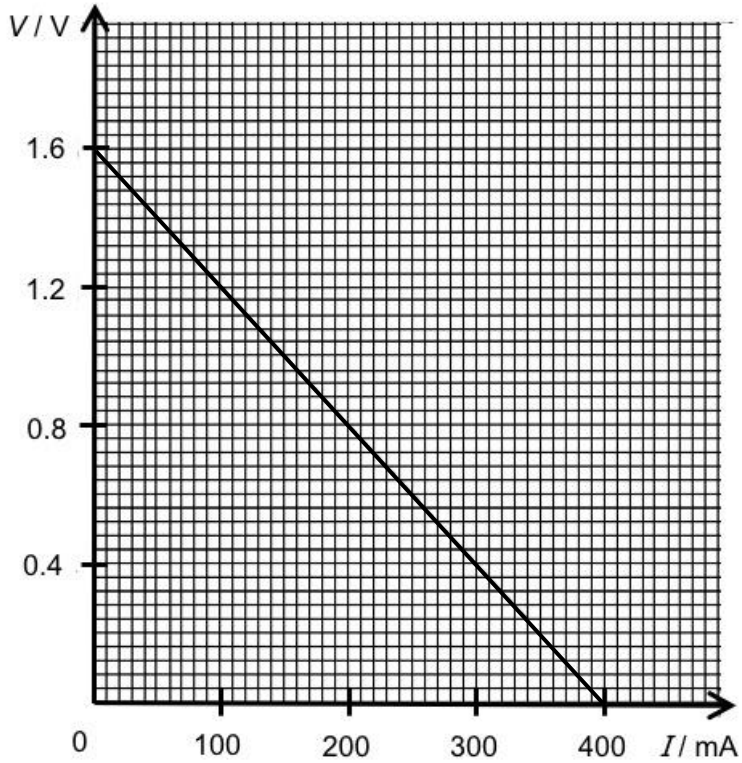

Fig. 8.2

(i) Deduce from Fig. 8.2 the e.m.f. *E* and the internal resistance *r* of the cell.

E = …….. V

r = …….. Ω [3]

(ii) J is placed at the position such that **maximum power** is delivered from the cell to the wire JN.

Determine the potential at J and N.

potential at J = ……V

potential at N = ……V [3]

(iii) On Fig. 8.3 sketch the graph to show how electric potential varies with position along the wire JN. Label the vertical axis clearly. Explain your answer.

Fig. 8.3

.....

.....

.....

..... [3]

(iv) Suggest, with a reason, why the position of the jockey J has to be shifted, when the battery has been used for a prolong period of time for maximum power transfer.

.....

.....

..... [1]

(c) **(i)** Explain what is meant by an *electric field*.

.....

..... [1]

(ii) Draw the charge distribution and the electric field around the charged metal bodies for the following bodies

1. an isolated positively charged sphere A.

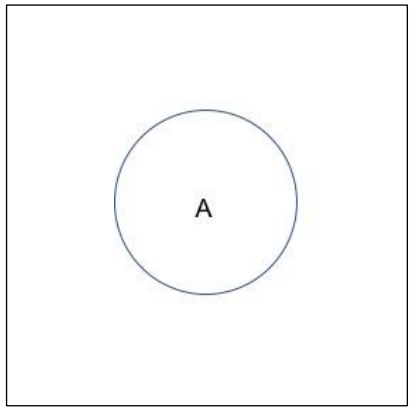

Fig. 8.1 [2]

2. When a neutral metal sphere B is brought close to the positively charged metal sphere A in **(c)(ii)1.**

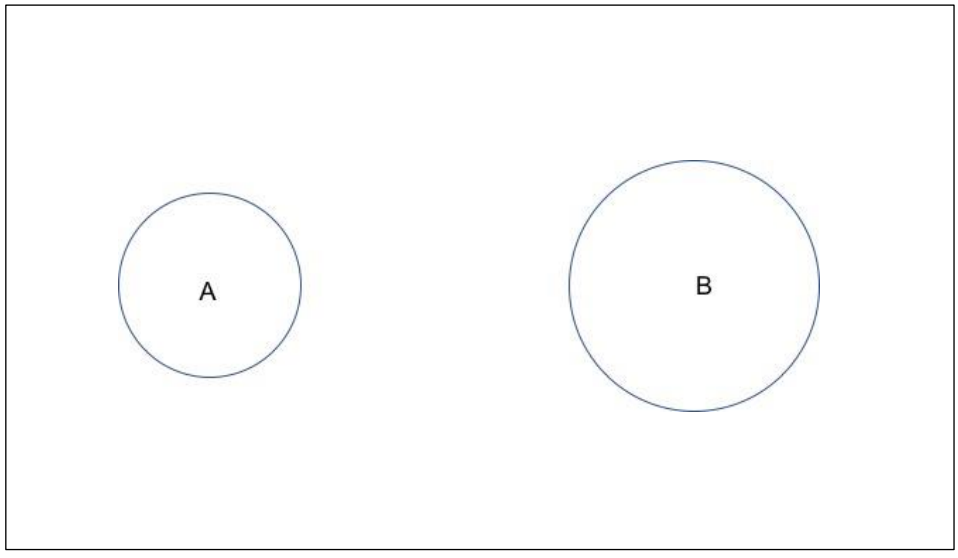

Fig. 8.2 [4]

- END OF PAPER -